ADE + FAB

ASSIGNMENT 4

CLUSTER ANALYSIS ON COMPANY FINANCIALS

- Machine Learning in Finance -

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INTRODUCTION

The main objective of this project is to apply cluster analysis to a dataset of 340 publicly traded companies in order to identify groups of firms with similar financial characteristics. The dataset includes a wide range of financial indicators such as stock returns, price volatility, profitability measures, liquidity ratios, and valuation metrics. These variables give us a snapshot of how companies are performing in the market and how financially stable or attractive they might be.

Instead of analysing companies one by one, clustering allows us to explore patterns in the data and group companies based on similarities in their financial profiles. This approach helps us to understand the structure within the dataset and to see whether certain types of companies tend to behave in similar ways. For example, we might discover clusters of high-growth firms, financially stable companies or businesses with high market volatility.

The purpose of this analysis is not only to group similar companies but also to gain insights into what defines those groups and how they might relate to real-world business categories, such as sectors or industries. This can be useful for investors, analysts, or companies themselves when making strategic decisions, comparing competitors, or building diversified portfolios.

DATA PREPARATION

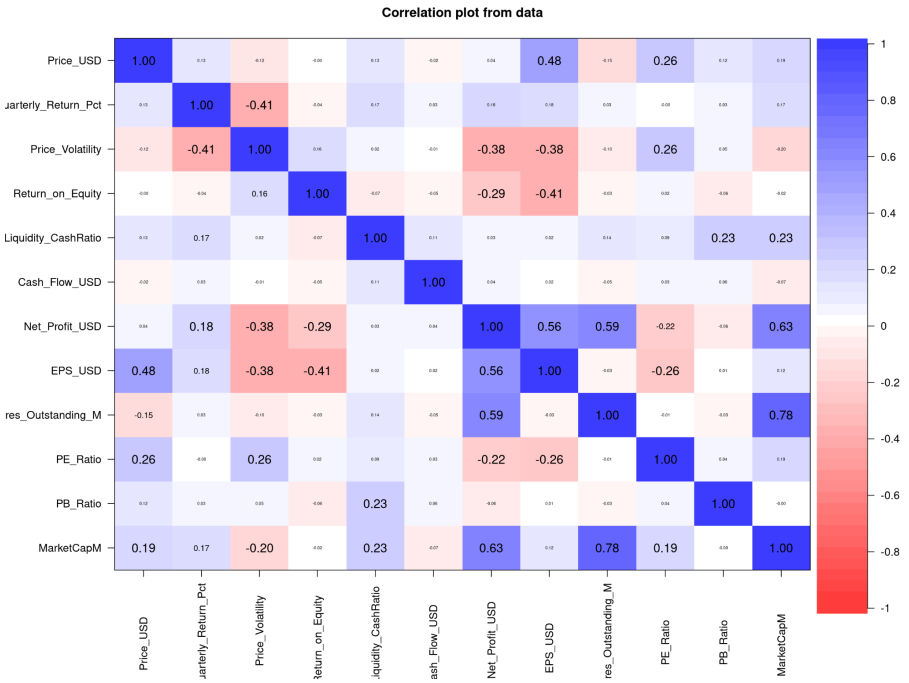
The original dataset included market and financial information for 340 companies. To prepare the data for clustering, we followed these steps:

Variable Selection: We selected the most relevant variables for the analysis, excluding categorical variables, as distances cannot be directly computed for them using standard clustering algorithms. We also excluded variables such as share price and shares outstanding individually, as they are highly correlated with market capitalization and offer limited insight on their own.

Rationale for Excluding Share Price and Shares Outstanding: Including share price or the number of shares independently can be misleading. A company might have a high number of cheap shares or a low number of expensive shares, so neither variable by itself accurately reflects the firm's size or market value. Only their product — market capitalization — gives a meaningful proxy of firm size (in terms of market value), which is more appropriate for clustering.

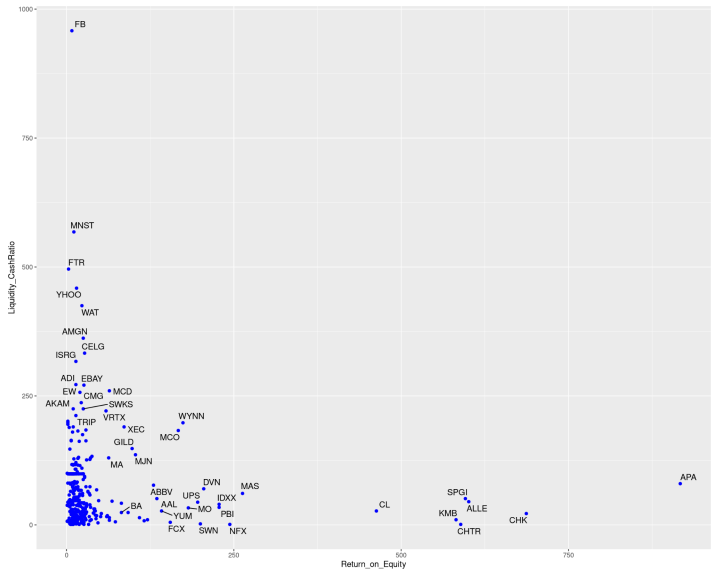
Treatment of Net Profit and EPS: We also noted that net profit is derived from earnings per share (EPS) multiplied by shares outstanding. While this raised the question of potential redundancy, we decided to retain EPS in the dataset. Since shares outstanding was already removed, excluding EPS as well would result in the complete loss of information related to firm-level profitability and dilution, which are valuable for characterizing companies in the clustering.

GRAPH 1: Correlation plot of numerical variables



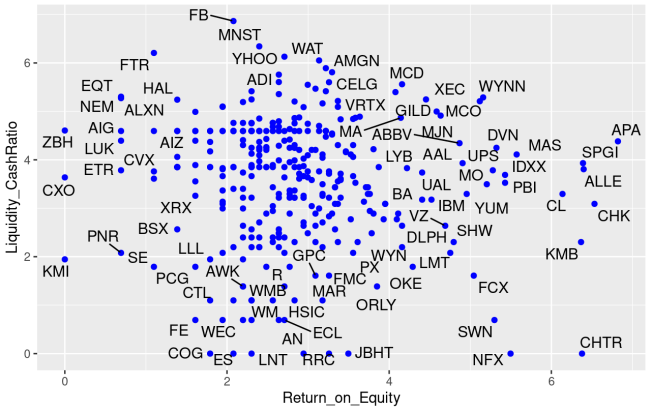
Some of the variables had skewed distributions or very large values, so we applied logarithmic transformations to reduce the effect of extreme values. Variables like “Liquidity_CashRatio”, “MarketCapM”, “PE_Ratio”, “Price_Volatility” and “Return_on_Equity” were transformed into their log because of their high asymmetry in the distribution, as can clearly be seen in the graphics. This logarithmic transformation helps us make the distributions symmetric.

GRAPH 2. Scatter plot of all companies comparing LCR and ROE



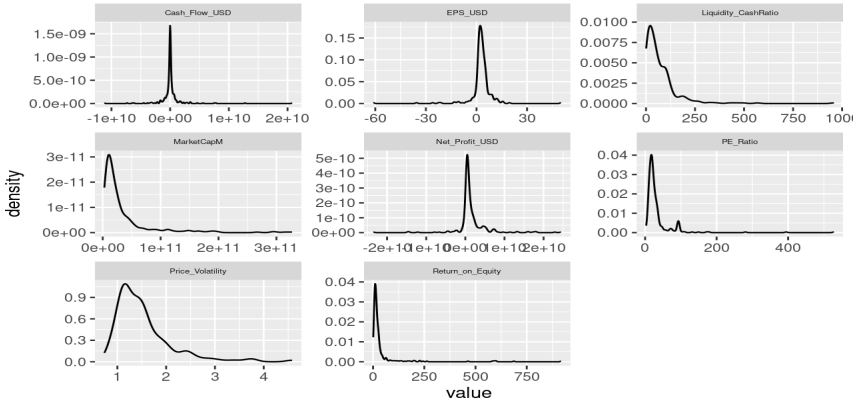
This scatterplot shows clearly the need for adapting the scales to logarithmic, as well as filtering out some outliers.

GRAPH 3. Scatter plot of all companies comparing LCR and ROE after logarithmic transformation



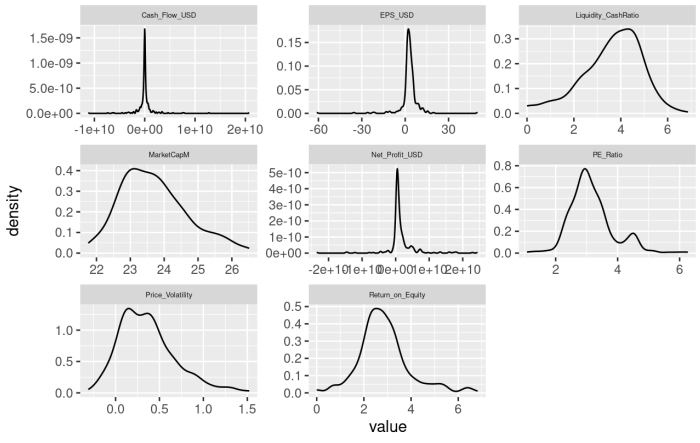
After applying the logarithm, it has a much better visualisation, with outliers playing a smaller role.

GRAPH 4: Distribution of numerical variables



This density plot shows the variables before the logarithmic transformation. The distributions are heavily skewed to the right, especially for variables like MarketCapM, Net_Profit_USD, and Return_on_Equity, which supports the decision to apply a log transformation to reduce the impact of extreme values and improve interpretability.

GRAPH 5: Distribution of numerical variables after logarithmic transformation



This set of density plots represents the variables after the logarithmic transformation. The distributions are visibly more symmetric and compact compared to their pre-log counterparts, especially for variables like MarketCapM, Return_on_Equity, and Liquidity_CashRatio. This transformation improved the normality and reduced the skewness, making the data more suitable for clustering analysis.

We ensured all variables were on the same scale by standardising them to a normal distribution with mean 0 and standard deviation 1. This is crucial in clustering to prevent variables with larger values from dominating the results. Standardisation puts all numeric variables on equal footing, making it easier to interpret their relative importance.

Before applying clustering methods, we conducted an initial exploration of the dataset to identify potential outliers. We focused on companies with standardised values exceeding four standard deviations from the mean. While a single outlier in one variable is typically not problematic, cases where a company shows outlier behaviour across multiple variables can significantly affect the analysis and are better excluded.

At first, the few outliers didn't seem to distort the cluster. But, as we kept on analysing the cluster, especially when using the hierarchical method, we noticed the emergence of little clusters formed of companies which were very different from the rest. These small and distinct groups interfered with the clarity and consistency of the cluster.

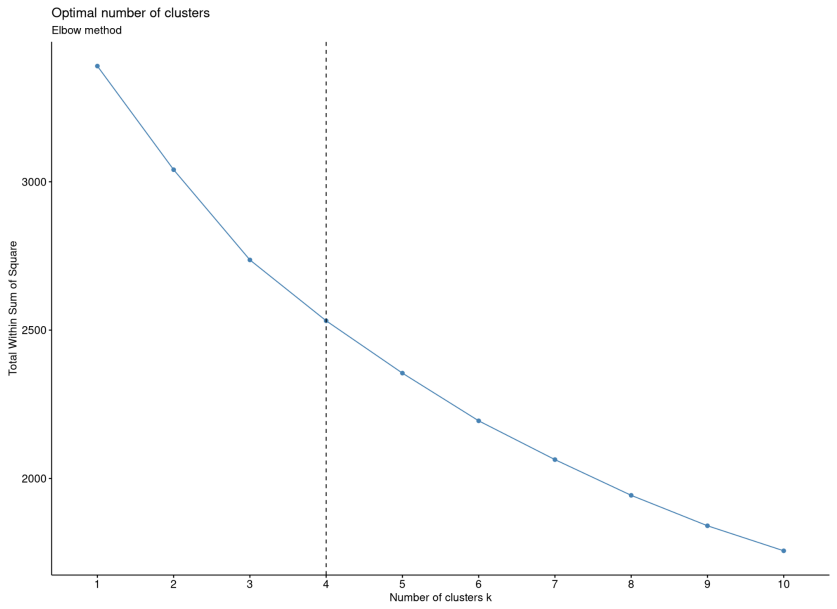
To address this, we adopted an iterative approach. We progressively filtered the dataset by removing specific companies or clusters that clearly skewed the results.
After performing every filtering, we reassessed the clustering performance and metrics quality. This iterative process lets us enhance the dataset, leading to more coherent and meaningful clusters.

METHODOLOGY

To identify the groups of companies with similar financial characteristics, we applied two different clustering methods: k-means clustering and hierarchical clustering using Ward’s method. These are both unsupervised learning techniques, which means the algorithm finds patterns in the data without us telling it what to look for. However, they work differently. Hierarchical clustering builds a tree-like structure (a dendrogram) by merging companies step by step based on their similarity, which is useful for visualising the clustering process and exploring different numbers of clusters by “cutting” the tree at different heights. In contrast, k-means assigns companies into a fixed number of clusters by minimising the variance within each cluster, which can be faster for large datasets but has some limitations. One of the main challenges with k-means is that it requires predefining the number of clusters (k), and it is sensitive to the random starting points, which can lead to different results in each run. Additionally, k-means does not always perform well with small datasets (as it is the case), as it tends to form clusters of similar sizes even when the data naturally has unbalanced groupings. This can cause k-means to create artificial splits or miss important patterns that hierarchical methods can capture more accurately in smaller datasets.

After preparing the data as mentioned in the previous section, we first applied hierarchical clustering using the Ward.D2 method with Euclidean distance. We tested different numbers of clusters by cutting the dendrogram at 4, 5, and 6 groups. We visualised the clusters using PCA-based plots to check how well-separated the clusters were, and we also compared different hierarchical methods (like complete linkage) to confirm that Ward's method performed better. During this process, we repeatedly checked the distribution of cluster sizes, identified small and unusual clusters, and removed problematic companies when necessary to improve the results.

Elbow rule for the first ward clustering



In parallel, we applied k-means clustering focused on solutions with 4, 5, and 6 clusters. We ran the algorithm multiple times with different random starting points to make sure the results were stable. Then, we visualised the clusters using PCA (Principal Component Analysis) plots, which help us understand how well-separated the groups are in a 2D space.

To compare both clustering methods, we used internal validation metrics that help measure how good and consistent the clusters are across methods. Once we saw which one was better, we created cross-tabulations between the groupings from k-means and hierarchical clustering. Since k-means clusters can be randomly labelled, we used an algorithm to match the clusters from k-means and hierarchical clustering based on similarity in metrics. We also used heatmaps to visualise the overlap between both methods, checking how many companies were grouped together in the same cluster across both approaches. We spotted a problematic group, so we isolated it and compared it to the 2 possible clusters that it could belong to. More specifically, we compared the group labelled as 2 by the K-means and 1 by the Ward to the groups labelled as 1 by the 2 methods and as 2 by the 2 methods. To do so, we compared the boxplots of the 3 groups and performed ANOVAs to check whether there were significant differences across the groups. Also, we checked whether the sectors were spread differently across those clusters, or if there were no differences.

To support the selection of the optimal clustering approach, we also computed internal validation metrics such as connectivity, Dunn index, and Silhouette score, which provided complementary insight into the compactness and separation of the clusters. On the one hand, connectivity measures

how well each observation is connected to its nearest neighbours, especially those in other clusters. Lower connectivity values indicate better clustering, as observations are more strongly connected within their own cluster and weakly connected to others. On the other hand, the Dunn Index evaluates the ratio between the minimum distance between clusters (i.e., how far apart clusters are) and the maximum intra-cluster distance (i.e., how large or dispersed a cluster is). A higher Dunn Index suggests more compact and well-separated clusters. Finally, the Silhouette score assesses how similar each observation is to its own cluster compared to the nearest other cluster. It ranges from -1 to 1, where higher values indicate that observations are well matched to their own cluster and clearly distinct from neighbouring clusters.

Next, we explored how the resulting clusters were related to categorical variables like Sector and Industry Group. We calculated the percentage of companies from each sector or industry in every cluster and created tables to summarise these relationships. This helped us see if certain clusters were dominated by specific sectors or if there was a mix of industries in each group.

To get a better understanding of what made each cluster different, we created logistic regression models for each cluster. These models helped us identify which financial variables had a significant impact on the likelihood of a company being in a specific group. At first, we used the p-values from the models to select the most relevant variables, using the ones present as significant in more than one model. However, in the end, we decided to rely on ANOVA tests to choose the variables, removing the only variable that did not show significant differences between clusters. Once we had those, we ran clustering again using only the significant variables to see if that would give us cleaner or more meaningful groups. We repeated hierarchical clustering using this reduced set of variables, and once again evaluated the number of clusters using the elbow method and PCA visualisation of the resulting clusters.

We even tried combining both clustering methods. First, we performed k-means clustering with a larger number of clusters (e.g., 8), and then applied hierarchical clustering on the centres of those clusters using Ward's method. This allowed us to summarise and simplify the k-means output and group similar clusters together. We visualised this hybrid approach with dendrograms and cluster profiles, which gave us another layer of insight into how companies are grouped.

Finally, we created the same number of clusters as sectors, and looked if they matched, as well as comparing the standardised profiles of the sectors to see if they were stable across all metrics.

Throughout the project, we used different visualisations like density plots, PCA plots, dendrograms, and line graphs of cluster profiles to better understand and explain our results. Each step helped us refine the analysis and build a more accurate and interpretable model for grouping companies based on their financial data.

RESULTS

This section presents the clustering outcomes obtained using different methods (Ward and k-means), datasets (full, filtered, and based on significant variables), and evaluation metrics. Visual

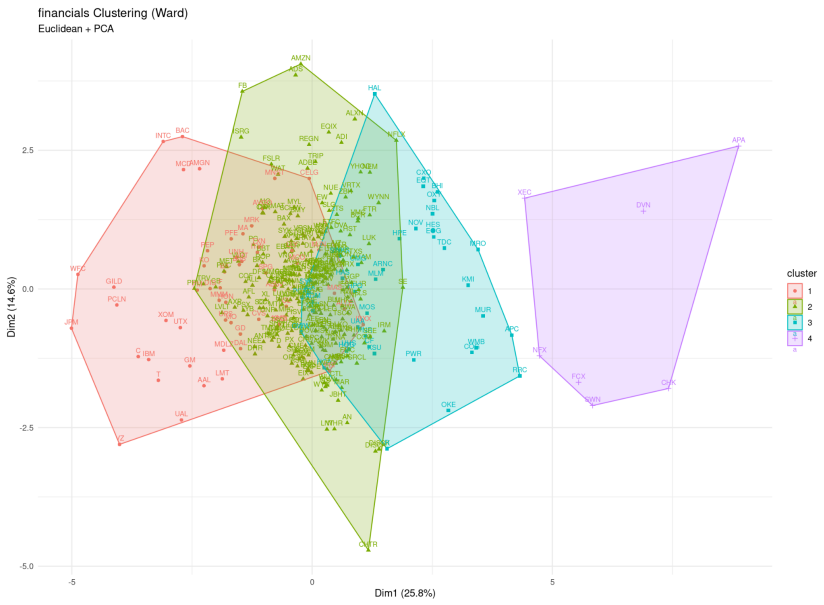
representations using PCA and internal validation scores (Silhouette, Dunn, and connectivity) are used to compare the structure, quality, and interpretability of each clustering approach.

TABLE 1: Hierarchical and non-hierarchical clustering methods comparison on different databases

	Full data (340)		Filtered data (333)			too filtered database (325)		Significant variables (333)	
	Hierarchical	K-Means	Hierarchical	K-Means	Mixed	Hierarchical	K-Means	Hierarchical	K-Means
Connectivity	9,6202	121,0282	10,2159	12,2698	NULL	9,4869	31,6663	10,2559	12,2698
Dunn	0,3469	0,048	0,4029	0,2241	0,1274628	0,3549	0,1125	0,2241	0,4029
Silhoutte	0,5561	0,2118	0,5854	0,4915	0,3410792	0,596	0,3861	0,4029	0,4915
Distribution	237, 55, 41, 7		107, 100, 94, 32	141, 107, 52, 33	299,23,9,2	208, 63, 38, 16		172, 99, 41, 21	

The table shown summarises the internal validation metrics (Connectivity, Dunn index, and Silhouette coefficient) across various dataset versions and clustering methods. It shows that the hierarchical method consistently outperforms K-Means, particularly in terms of connectivity and silhouette score. The best overall results appear in the filtered data using hierarchical clustering (e.g., a silhouette of 0.5854 and Dunn of 0.4029), indicating better-defined and more compact clusters after removing outliers. The titles are based on the database, being full the first one, filtered the one with the 7 outliers removed, the too filtered the one after the second filter and the last one the one performed with only significant variables in the filtered data.

GRAPH 6: First PCA projection with 4 clusters (Ward's method)

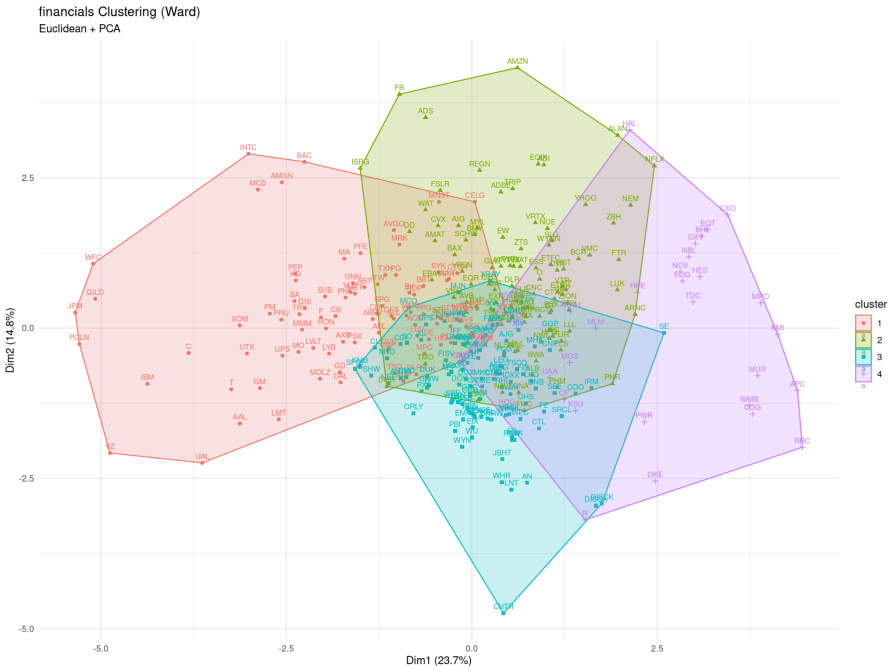


Graph 6 displays the result of a Principal Component Analysis (PCA) projection of the 340 firms onto the first two principal components, which explain 25.8% and 14.6% of the variance, respectively. The firms are grouped into four clusters identified using Ward's hierarchical clustering method. As we can see, Cluster 1 (in red) occupies the left region of the plot and includes firms with more extreme positions on both dimensions. Cluster 2 (in green) appears more centralised and dense, spreading across the origin and overlapping partially with Clusters 1 and 3. Cluster 3 (in blue) is positioned to

the right of the centre and appears more compact than the others. Cluster 4 (in purple) is clearly separated from the others, located in the far right area of the plot.

These observations are supported by the internal clustering validation metrics reported in Table 1. For this configuration (Ward on 340 firms), the Silhouette coefficient is 0.5561, the Dunn index is 0.3469, and the connectivity value is 9.6206. In comparison, k-means on the same dataset shows notably poorer results, with a Silhouette of 0.2118, a Dunn index of 0.048, and connectivity of 121.03. Despite the relatively good performance of Ward's method here, both solutions are outperformed by the configurations applied to the filtered dataset, which yield stronger validation metrics. Therefore, neither of these full-dataset clusterings is selected as the definitive result.

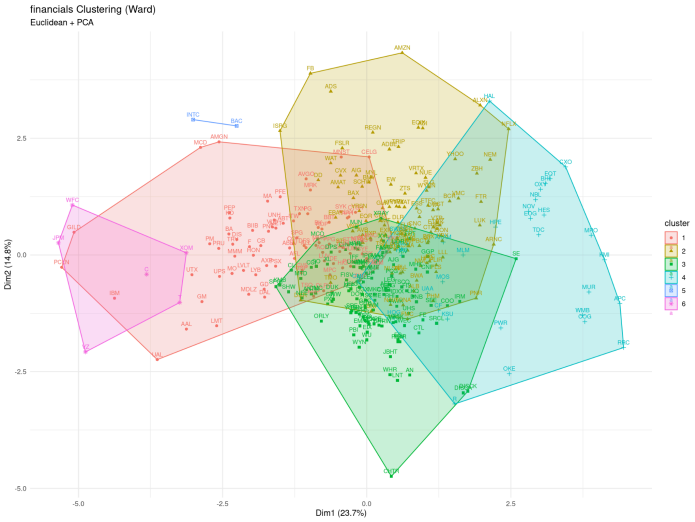
GRAPH 7: PCA projection with 4 clusters (Ward's method) after removing Graph 6's cluster 4



The new clusters, obtained from the filtered dataset of 333 firms, differ in compactness and internal cohesion compared to the original configuration. Cluster 1 appears more dispersed, while Clusters 2 and 3 are denser and show tighter groupings. The removal of the excluded firms contributes to clearer separation between groups and enhances the visual clarity of the PCA projection. As a result, the updated clustering structure allows for a more focused and interpretable analysis of the remaining firms, with better-defined boundaries and reduced overlap between clusters.

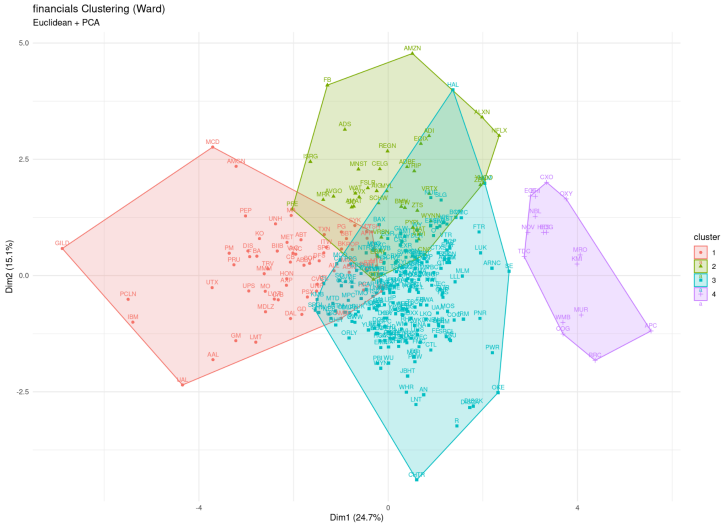
These improvements are also supported by the internal validation metrics shown in Table 1. In this configuration, Ward's method achieves the highest Silhouette coefficient (0.5854) and Dunn index (0.4029) among all tested approaches, along with a low connectivity value (10.2559). Furthermore, k-means also performs significantly better on the filtered dataset, with a Silhouette of 0.4915, Dunn index of 0.2241, and connectivity of 27.8586, representing a major improvement over its performance on the full dataset. These results confirm that the clustering applied to the filtered dataset, regardless of method, offers superior structure and quality.

GRAPH 8: PCA projection with 6 clusters (Ward's method) after removing Graph 6's cluster 4



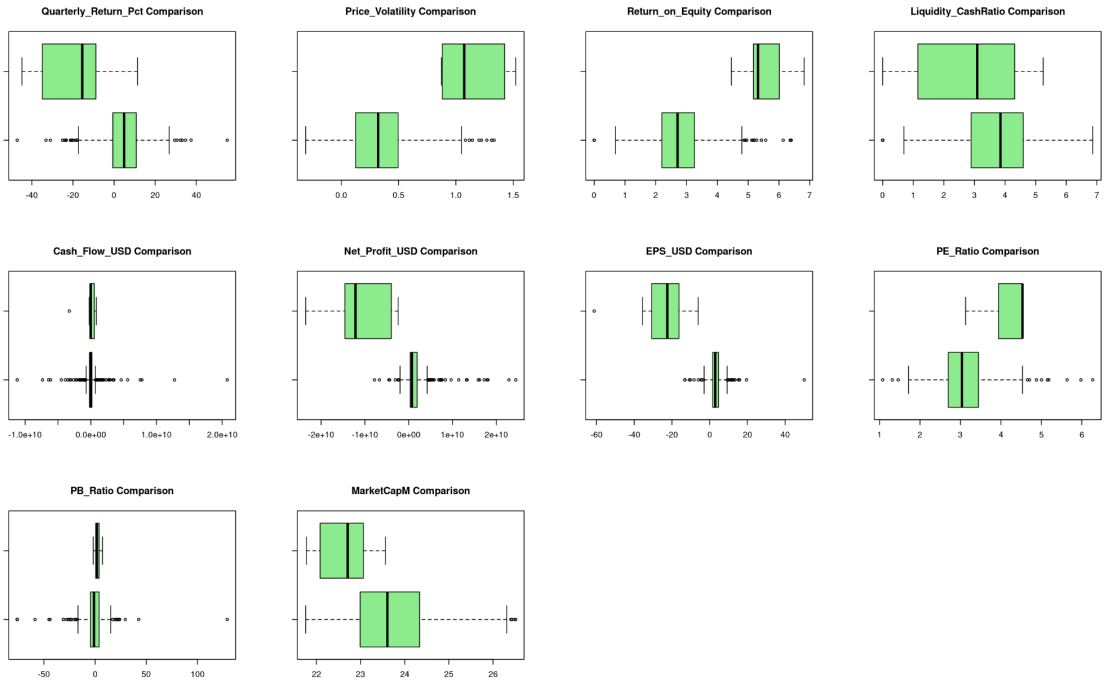
Compared to the previous four-cluster configuration, the introduction of two additional clusters (Clusters 5 and 6) results in a more granular segmentation of the data. These two clusters consist of relatively few firms and appear as small, compact groupings separated from the denser central structure. Their formation suggests the presence of distinct patterns within a limited number of observations, which may indicate that these clusters are capturing small subgroups or potential outliers. However, the internal validation metrics for the six-cluster solution indicate weaker performance compared to the best four-cluster configurations discussed in Table 1. For the hierarchical method with six clusters, the Silhouette coefficient is 0.4687, the Dunn index is 0.2284, and connectivity is 24.7456, all lower than those obtained using Ward with 4 clusters (Silhouette 0.5854, Dunn 0.4029, connectivity 10.2559). Similarly, for k-means with six clusters, the Silhouette (0.1741), Dunn (0.0423), and connectivity (179.3770) values are substantially worse than the four-cluster k-means configuration on filtered data. These results suggest that, despite the added segmentation, the six-cluster solution does not improve the overall clustering quality and is therefore not preferred as the final model.

GRAPH 8.1: PCA projection with 4 clusters (Ward's method) after removing Graph 8's cluster 5 & 6



After performing the cluster analysis again we find more separated clusters, however as can be seen in table 1, it has some worse metrics as well as more differences with k-means, so we do not choose it as the final model.

GRAPH 9: Comparison of boxplots of cluster 4 (upper box) and the rest of the database

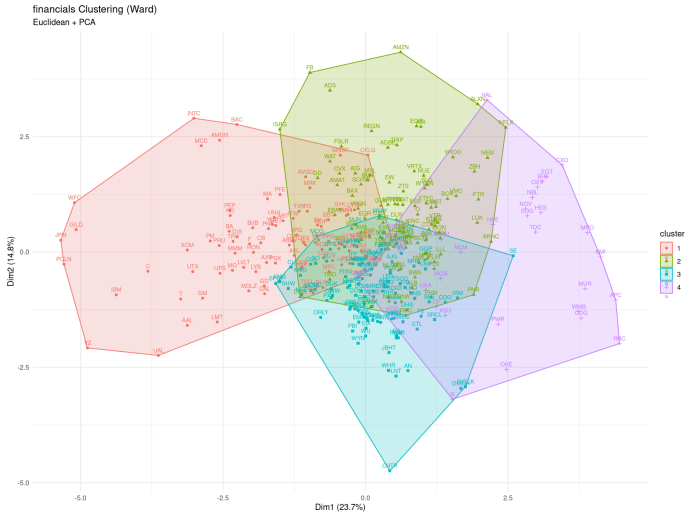


This graphic contrasts the eliminated 4th cluster to the rest of the database. Across most variables, the boxplots reveal noticeable differences in central tendency and dispersion between Cluster 4 and the remaining firms. For example, Cluster 4 shows a higher median in variables such as Quarterly Return Percentage, Return on Equity, and Liquidity Ratio. In contrast, the dispersion is generally narrower for variables like EPS, Net Profit, and Cash Flow, where the boxes are more compact and display fewer extreme values in Cluster 5 compared to the rest of the database.

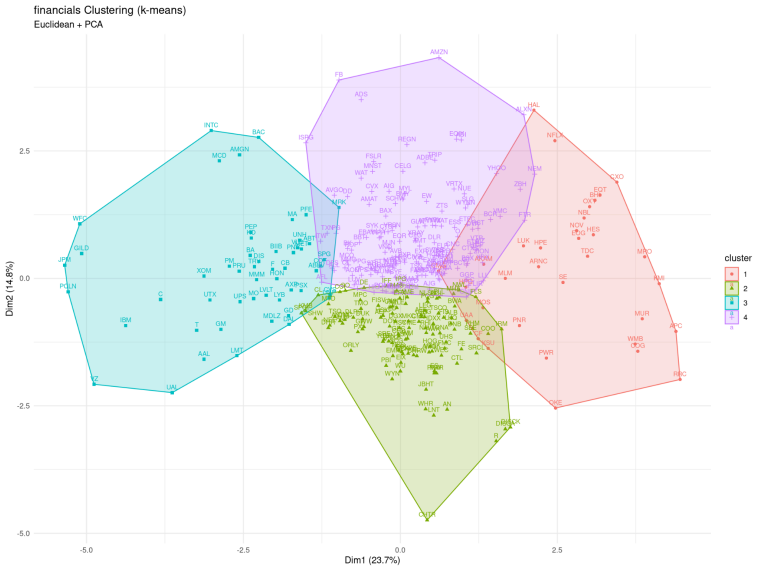
In several cases (e.g., MarketCapM and PE Ratio), the interquartile ranges of Cluster 4 are visibly distinct from those of the rest, indicating differences in the spread and scale of these features. Outliers are present in both groups across multiple variables, although they tend to be more prominent and numerous in the non-cluster firms (as it is logical due to the higher size of the database).

This graphical summary provides a visual overview of how Cluster 4 firms differ in terms of financial metrics distribution, as compared to the rest of the sample.

GRAPH 6 for comparison



GRAPH 10: PCA projection with 4 clusters (K-means) after removing Graph 6's cluster 4

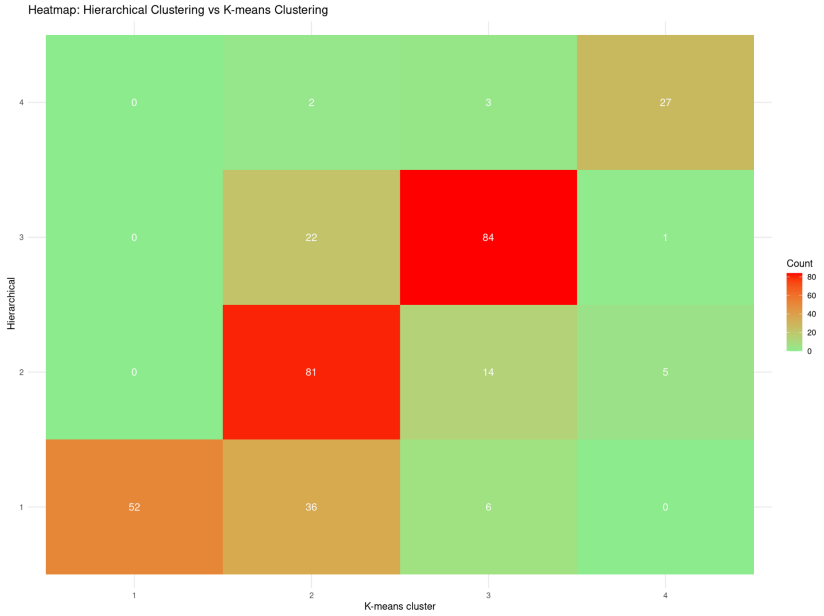


The two graphs display the same filtered dataset (333 firms) clustered into four groups using two different algorithms: Ward's hierarchical clustering (top) and k-means clustering (bottom), both projected onto the first two principal components. While both methods produce comparable global segmentation, notable differences arise in the shape and cohesion of the clusters.

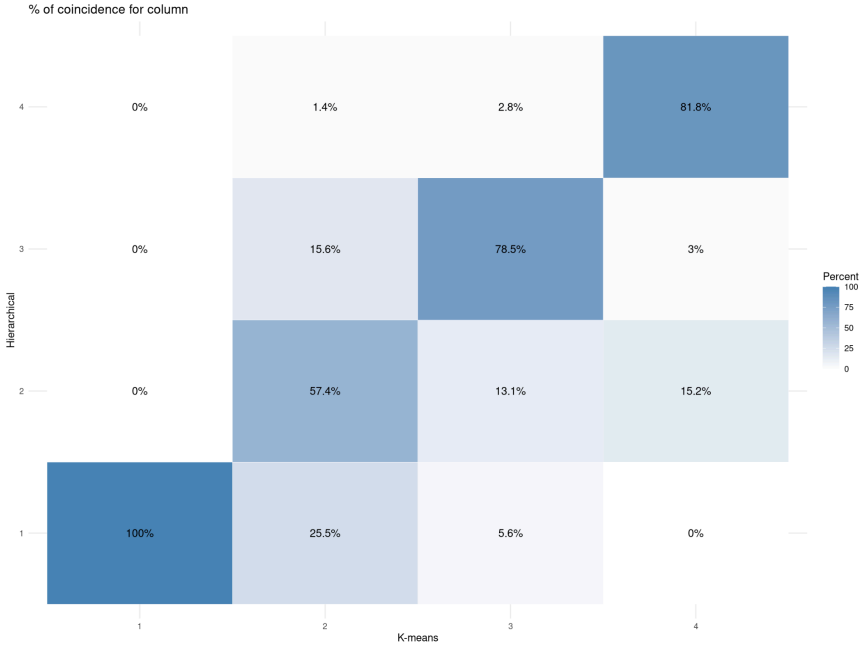
Ward's method yields clusters with more irregular and adaptive boundaries, better capturing local density and variations in firm distribution. In contrast, k-means produces clusters that are more symmetric and geometrically regular due to its centroid-based nature. This results in some firms being grouped differently between the two methods, particularly around cluster borders and in denser regions of the plot. Additionally, the central area in the k-means solution appears more tightly structured, while Ward's method shows a more nuanced partitioning of that space.

These visual differences are supported by the internal validation metrics reported in Table 1. On the filtered dataset, Ward's method again outperforms k-means, with a Silhouette coefficient of 0.5854 versus 0.4915, a Dunn index of 0.4029 versus 0.2241, and lower connectivity (10.2559 vs. 27.8586). These results indicate that the hierarchical approach offers better cluster compactness, separation, and overall quality, even when applied to the reduced dataset.

GRAPH 11: Heatmap comparison of Ward and K-means by individuals



GRAPH 12: Heatmap comparison of Ward and Kmeans by individuals as % of k-means



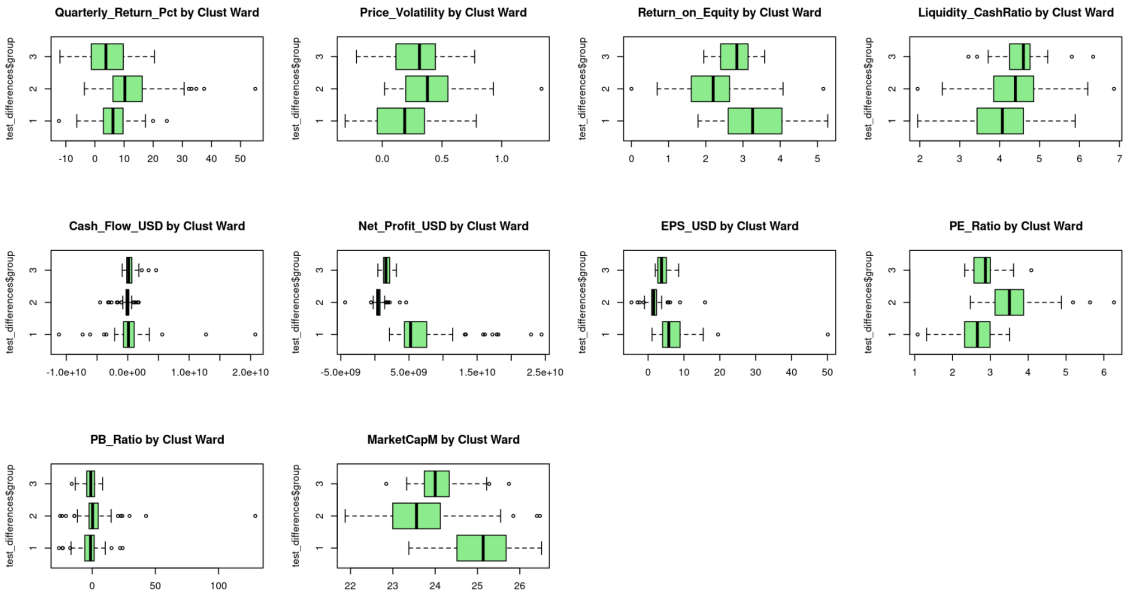
The comparison heatmaps between Ward's hierarchical clustering and k-means clustering reveal substantial alignment between the two methods, particularly in the core clusters. As shown in Graph 11, the strongest agreement occurs between Cluster 3 in both methods, with 84 shared assignments, followed by Cluster 2, where 81 firms are matched. These cells, highlighted in red, indicate highly stable groups where both algorithms converge, suggesting well-separated and internally cohesive profiles.

Since k-means assigns cluster labels arbitrarily, the numeric labels do not inherently match those from hierarchical clustering. To enable a direct comparison, we realigned the k-means labels by solving a linear sum assignment problem (LSAP) using the confusion matrix between the two methods.

Notably, Ward's Cluster 1 overlaps with k-means Clusters 1 and 2, indicating ambiguity in the boundary definition of this group. This partial mismatch is visualised more clearly in Graph 12, which presents the cluster correspondence as a percentage of each k-means cluster. For example, k-means Cluster 3 aligns 78.5% with Ward's Cluster 3, and k-means Cluster 4 matches 81.8% with Ward's Cluster 4. In contrast, k-means Cluster 2 matches only 57.4% with Ward's Cluster 2, while 25.5% of its firms are grouped into Ward's Cluster 1, reflecting a meaningful divergence in how these firms are interpreted.

Due to this partial disagreement, particularly across cluster intersections (1,1), (2,2), and (1,2), we conducted further analysis focused on these three groupings. This allowed us to better understand where both clustering methods align and where they differ in their classification of similar firm profiles.

GRAPH 13: Boxplot comparison of the 3 previously mentioned groups



Graph 13 presents boxplots comparing the three previously identified groups: (1,1), (2,2), and (1,2), across ten financial variables. These groups represent combinations of cluster assignments from Ward's and k-means methods, where (1,1) and (2,2) indicate alignment between both methods, and (1,2) reflects a disagreement.

Visual inspection reveals that the distributions across several variables differ across the three groups. The (1,2) group tends to show more dispersion and greater interquartile ranges in variables such as Quarterly_Return_Pct, Price_Volatility, and Return_on_Equity. In contrast, the (1,1) and (2,2) groups generally exhibit tighter boxplots, suggesting more concentrated distributions.

Differences in median values are also noticeable in variables such as Liquidity_CashRatio, PE_Ratio, and MarketCapM, where the central tendencies appear distinct across groups. For variables like Net_Profit_USD, Cash_Flow_USD, and EPS_USD, the (2,2) group displays lower overall dispersion, while the (1,2) group shows a wider spread and more outliers.

These patterns support the decision to further analyse these three groups, as their financial characteristics are visibly different in multiple dimensions.

TABLE 2: Sector distribution in the previously mentioned groups, as column %

Sector	Group 1	Group 2	Group 3
Health Care	0,17	0,08	0,12
Industrials	0,13	0,15	0,21
Consumer Discretionary	0,12	0,11	0,12
Financials	0,12	0,16	0,14
Energy	0,11	0,1	0,03
Utilities	0,09	0,05	0,08
Real State	0,08	0,08	0,08
Information technology	0,07	0,13	0,07
Consumer Staples	0,05	0,06	0,07
Materials	0,03	0,08	0,06
Telecommunications	0,03	0,01	0,01

The sectoral breakdown of the three groups (really significant by the chi-sq) reveals differences in their economic composition that may help explain some of the variations observed in the financial variables. Group 1, which aligns across both clustering methods (1,1), shows higher representation from the Health Care (17%), Industrials (13%), and Consumer Discretionary (12%) sectors. In contrast, Group 2 (2,2) stands out for its relatively stronger presence in Financials (16%) and Information Technology (13%).

Group 3, corresponding to the divergent case (1,2), differs notably in its sector mix. It has a disproportionately high share of Industrials (21%) and a significantly lower presence of Energy firms (3%) compared to the other two groups. These differences may contribute to the distinct financial profiles observed in the boxplots, particularly in variables like return on equity, volatility, and profitability.

The data suggest that sectoral composition plays a role in shaping the clustering structure and may partly explain why the two methods diverge in their classification of some firms.

TABLE 3: P-values after performing anova with all variables and the cluster they belong to, at a 5% significance level

Variable	Significancy group 1	Significancy group 2
Quarterly_Return_Pct	FALSE	TRUE
Price_Volatility	TRUE	TRUE
Return_on_Equity	TRUE	TRUE
Liquidity_CashRatio	TRUE	FALSE
Cash_Flow_USD	FALSE	TRUE
Net_Profit_USD	TRUE	TRUE
EPS_USD	TRUE	TRUE
PE_Ratio	TRUE	TRUE
PB_Ratio	FALSE	FALSE
MarketCapM	TRUE	TRUE

To assess the statistical differences between the aligned and divergent groups, ANOVA tests were performed comparing group (1,1) with (1,2), and group (2,2) with (1,2), with significance determined at the 0.05 level. The results are summarised in the table above.

For the comparison between groups (1,1) and (1,2), significant differences were found in 7 out of the 10 variables, including all but Quarterly_Return_Pct, Cash_Flow_USD, and PB_Ratio.

The comparison between groups (2,2) and (1,2) revealed an even clearer distinction, with 9 out of 10 variables reaching significance. Only Liquidity_CashRatio did not differ significantly in this case.

These results confirm that group (1,2), composed of firms classified differently by Ward and k-means, is consistently distinct from both aligned groups. Among the two aligned groups, group (1,1) exhibits fewer differences when compared with the divergent group, suggesting that it may represent a more internally coherent or stable segment under both clustering approaches, and shows that Ward remains a better choice.

Now, for the analysis of the variables' patterns in the clusters we have created:

TABLE 4: Sector distribution by cluster, in % of each cluster

Sector	Cluster 1	Cluster 2	Cluster 3	Cluster 4
Financials	28	14	8	0
Industrials	21	9	19	12
Health Care	15	16	9	0
Consumer Staples	10	2	7	0
Information Technology	7	19	3	12
Consumer Discretionary	6	13	16	12
Energy	5	1	1	53
Telecommunications services	3	1	1	0
Material	2	7	7	9
Real State	1	16	9	0
Utilities	1	2	20	0

The table above presents the sectoral composition of each cluster in percentage terms (really significant by the chi-sq). Clear differences in industry representation can be observed across the four groups. Cluster 4 is highly concentrated, with 53% of its members belonging to the Energy sector, making it the most homogeneous cluster. In contrast, Cluster 1 displays a more diversified sectoral profile, with notable representation from Financials (28%), Industrials (21%), and Health Care (15%), among others.

Cluster 2 stands out for its relatively high proportions of Information Technology (19%) and Health Care (16%), alongside Real Estate and Financials. Cluster 3 includes a significant share of Utilities (20%), as well as meaningful representation from Consumer Discretionary (16%) and Industrials (19%).

These distributions suggest that the clustering process effectively groups firms with shared sector characteristics in some cases, while maintaining more heterogeneous compositions in others. The emergence of a distinct Energy-dominated cluster, for example, indicates strong underlying commonalities within that group.

Logistic regression provided a quantitative link between sector affiliation and cluster membership. The models showed statistically significant predictive power from financial ratios and categorical variables such as sector. This further confirms that the clusters have external validity—that is, they correspond to real-world distinctions that go beyond the numeric dataset. Particularly, variables like ROE and MarketCapM were consistent and powerful predictors, which affirms their importance not

just in theoretical finance but in practical company analysis and segmentation. It is relevant to mention that regressions performed with the variable Sector took Consumer Discretionary as base level, but changes on the level did not significantly affect the significance of this variable. Furthermore, accuracy has some limitations, as it is calculated without train and test splitting, due to the small size of the sample and knowing that clusters are not real "predictions" as it is an unsupervised model. Furthermore, intercept doesn't have a clear interpretation, as some variables can't be 0. Finally, the regressions performed with "Industry Sector", as the great amount of variables led to unreliable results.

TABLE 5: Logistic regression results and accuracy for the first cluster, with numerical variables only

Model 1	Estimate	Pr(>[z])	Significancy
(Intercept)	-83,89	7.27e-07	***
Quarterly_Return_Pct	0,00	0,77	
Price_Volatility	-0,09	0,38	
Return_on_Equity	0,59	0,05	.
Liquidity_CashRatio	1,37	2.04e-05	***
Cash_Flow_USD	0,00	2.68e-05	***
Net_Profit_USD	0,00	0,52	
EPS_USD	0,25	0,02	*
PE_Ratio	-5,66	1.74e-07	***
PB_Ratio	0,00	0,94	
MarketCapM	3,82	9.76e-07	***
Residual variance	95,203		
AIC	117,2		
Accuracy	94,59%		

Model 1 shows that Liquidity_CashRatio, MarketCapM, and PE_Ratio have strong effects on cluster membership, with positive effects for the first two and a negative effect for the latter. EPS_USD and Return_on_Equity also contribute positively, though to a lesser extent. The rest of the variables do not show relevant effects in this model, being most of them close to 0.

The model achieves high accuracy (94.59%), a low AIC (117.2), and a residual deviance of 95.203, indicating strong model fit and predictive performance.

TABLE 6: Logistic regression results and accuracy for the first cluster, with numerical variables and sector

Model 1.1	Estimate	Pr(>[z])	Significancy
(Intercept)	-83,75	7.51e-06	***
Quarterly_Return_Pct	-0,01	0,07	
Price_Volatility	-1,98	0,02	
Return_on_Equity	0,51	0,01	
Liquidity_CashRatio	1,74	0,00	***
Cash_Flow_USD	0,00	5.23e-05	***
Net_Profit_USD	0,00	0,03	
EPS_USD	0,18	0,01	
PE_Ratio	-5,98	1.80e-06	
PB_Ratio	-0,01	0,08	
MarketCapM	3,83	8.25e-06	***
SectorConsumer Staples	0,10	0,09	
SectorEnergy	-0,31	0,09	
SectorFinancials	-1,01	0,05	
SectorHealth Care	-0,20	0,09	
SectorIndustrials	1,11	0,04	
SectorInformation Technology	-2,38	0,02	
SectorMaterials	0,62	0,07	
SectorReal Estate	-0,82	0,07	
SectorTelecommunications Services	0,00	0,10	
SectorUtilities	-1,83	0,34	
Residual deviance	86,358		
AIC	128,36		
Accuracy	93,69%		

Model 1.1, which incorporates sector variables alongside financial predictors, confirms the strong effects of Liquidity_CashRatio, Cash_Flow_USD, and MarketCapM, all highly significant. Although the coefficient for Cash_Flow_USD is close to zero, this is likely due to its large scale rather than a lack of predictive power. Among sector variables, Industrials and Information Technology show significant effects, while others are less conclusive.

The model maintains strong performance, with 93.69% accuracy, AIC of 128.36, and residual deviance of 86.36, confirming the added explanatory value of sector information.

TABLE 7: Logistic regression results and accuracy for the second cluster, with numerical variables only

Model 2	Estimate	Pr(>[z])	Significancy
(Intercept)	15,5400	0,015	*
Quarterly_Return_Pct	0,2182	1.98e-12	***
Price_Volatility	2,5610	0,003	**
Return_on_Equity	-2,2810	1.32e-09	***
Liquidity_CashRatio	0,4449	0,012	*
Cash_Flow_USD	-4,36E-10	0,008	**
Net_Profit_USD	-0,0168	0,226	
EPS_USD	-0,0829	0,323	
PE_Ratio	0,5415	0,154	
PB_Ratio	0,0337	0,011	*
MarketCapM	-0,6675	0,022	*
Residual variance	177,77		
AIC	199,77		

Model 2 predicts membership in Cluster 2 using only financial variables. The most significant predictors are Quarterly_Return_Pct, Price_Volatility and Return_on_Equity, indicating that this cluster is strongly influenced by market behaviour and performance volatility. Liquidity_CashRatio and Cash_Flow_USD are also significant, but contribute in opposite directions: higher liquidity increases the likelihood of cluster membership, while higher cash flow reduces it. Additionally, PB_Ratio and MarketCapM show smaller but significant effects.

In contrast, Net_Profit_USD, EPS_USD, and PE_Ratio are not significant in this model. The overall model fit is lower compared to previous models, with a residual variance of 177.77 and an AIC of 199.77, suggesting moderate predictive capacity.

TABLE 8: Logistic regression results and accuracy for the second cluster, with numerical variables and sector

Model 2.1	Estimate	Pr(>[z])	Significancy
(Intercept)	5,77	0,507	
Quarterly_Return_Pct	0,25	2.38e-08	***
Price_Volatility	5,03	0,000	***
Return_on_Equity	-3,35	4.30e-07	***
Liquidity_CashRatio	0,30	0,258	
Cash_Flow_USD	-6,55E-10	0,019	*
Net_Profit_USD	-4,32E-10	0,057	
EPS_USD	-0,30	0,022	*
PE_Ratio	0,58	0,315	
PB_Ratio	0,06	0,006	**
MarketCapM	-0,07	0,866	
SectorConsumer Staples	-3,50	0,020	*
SectorEnergy	-10,77	4.24e-05	***
SectorFinancials	-0,90	0,433	
SectorHealth Care	-1,47	0,160	
SectorIndustrials	-1,28	0,190	
SectorInformation Technology	-0,07	0,955	
SectorMaterials	-1,71	0,134	
SectorReal Estate	-0,52	0,599	
SectorTelecommunications Services	-4,19	0,330	
SectorUtilities	-2,52	0,781	
Residual variance	120,43		
AIC	162,43		
Accuracy	92,49%		

Model 2.1 combines financial and sector variables to predict membership in Cluster 2. Quarterly_Return_Pct, Price_Volatility, and Return_on_Equity remain highly significant ($p < 0.001$), confirming their strong influence. Cash_Flow_USD, EPS_USD, and PB_Ratio also contribute significantly. In contrast, MarketCapM, previously relevant in Model 2, loses significance here—possibly due to overlap with sector effects.

Among sectors, Energy shows a strong negative association with Cluster 2 (as they are only 1% of the individuals of the cluster), while Consumer Staples also has a significant negative effect ($p = 0.02$). Other sector variables are not statistically significant.

The model achieves 92.49% accuracy, with an AIC of 162.43 and residual variance of 120.43, indicating a slight decrease in fit compared to Model 2, but providing richer interpretability through sectoral structure.

TABLE 9: Logistic regression results and accuracy for the third cluster, with numerical variables only

Model 3	Estimate	Pr(>[z])	Significancy
(Intercept)	51,71	6.15e-08	***
Quarterly_Return_Pct	-0,06	0,023	*
Price_Volatility	-9,06	4.13e-12	***
Return_on_Equity	1,22	8.51e-06	***
Liquidity_CashRatio	-1,64	5.62e-10	***
Cash_Flow_USD	3,21E-10	0,197	
Net_Profit_USD	-8,03E-10	0,000	***
EPS_USD	0,12	0,037	*
PE_Ratio	-687,40	0,162	
PB_Ratio	-0,09	4.71e-05	***
MarketCapM	-1,92	2.61e-06	***
Residual variance	151,22		
AIC	173,22		

Model 3 evaluates the third cluster using only financial variables. The results show that firms with lower price volatility, lower liquidity, and smaller market capitalisation are more likely to belong to

this cluster. In contrast, higher return on equity and higher earnings per share increase the likelihood of cluster membership.

Notably, net profit, valuation ratios (PB), and short-term returns are also significant, but their negative direction suggests that this cluster is associated with firms that are financially conservative or undervalued, despite showing some signs of profitability.

Variables such as cash flow and PE ratio do not contribute significantly in this case. Although the model's residual variance (151.22) and AIC (173.22) are higher than in previous models, it still identifies a coherent profile of low-volatility, low-growth firms with selective strengths in profitability.

TABLE 10: Logistic regression results and accuracy for the third cluster, with numerical variables and sector

Model 3.1	Estimate	Pr(>[z])	Significancy
(Intercept)	57,12	1.11e-06	***
Quarterly_Return_Pct	-0,06	0,034	*
Price_Volatility	-8,65	3.96e-08	***
Return_on_Equity	1,70	2.59e-06	***
Liquidity_CashRatio	-1,60	3.58e-06	***
Cash_Flow_USD	1,76E-10	0,515	
Net_Profit_USD	-9,26E-10	0,001	**
EPS_USD	0,12	0,062	
PE_Ratio	-0,80	0,195	
PB_Ratio	-0,10	3.47e-05	***
MarketCapM	-2,24	9.84e-06	***
SectorConsumer Staples	1,45	0,255	
SectorEnergy	-1,13	0,557	
SectorFinancials	0,64	0,542	
SectorHealth Care	2,79	0,009	**
SectorIndustrials	-0,12	0,888	
SectorInformation Technology	0,11	0,946	
SectorMaterials	0,11	0,919	
SectorReal Estate	1,27	0,191	
SectorTelecommunications Services	1,63	0,582	
SectorUtilities	2,51	0,042	*
Residual variance	136,36		
AIC	178,36		
Accuracy	92,49%		

Model 3.1 combines financial variables and sector dummies to explain membership in Cluster 3. Several financial predictors — Price_Volatility, Return_on_Equity, Liquidity_CashRatio, Net_Profit_USD, PB_Ratio, and MarketCapM — remain highly significant, suggesting that this cluster is characterized by lower volatility, lower liquidity, lower valuation, and mixed profitability signals. Cash_Flow_USD, EPS_USD, and PE_Ratio are not significant.

Regarding sectoral variables, Health Care and Utilities are significant and positively associated with belonging to Cluster 3. This is consistent with the sector distribution percentages, where Utilities represent 20% of Cluster 3, and Health Care 9%, being industrial the highest concentration for that sector across all clusters. Meanwhile, non-significant sectors like Energy and Financials have minimal representation in Cluster 3 (both around or below 1%), supporting their lack of statistical effect in the model.

With 92.49% accuracy, AIC of 178.36, and residual variance of 136.36, this model offers improved interpretability over the financial-only version by highlighting that Cluster 3 not only reflects a

valuation-based financial profile but also shows moderate sectoral alignment—particularly in Health Care and Utilities.

TABLE 11: Logistic regression results and accuracy for the fourth cluster, with numerical variables only

Model 4	Estimate	Pr(>[z])	Significancy
(Intercept)	-0,04	0,998	
Quarterly_Return_Pct	-15,47	0,992	
Price_Volatility	384,80	0,993	
Return_on_Equity	55,81	0,993	
Liquidity_CashRatio	34,96	0,996	
Cash_Flow_USD	2,97E-08	0,997	
Net_Profit_USD	-2,94E-08	0,998	
EPS_USD	4,00	0,997	
PE_Ratio	44,35	0,997	
PB_Ratio	2,70	0,993	
MarketCapM	-15,18	0,999	
Residual variance	2,61E-07		
AIC	22		

This table displays the results for Model 4, which attempts to predict cluster assignment using financial variables. However, the model fails to identify any statistically significant predictors. All variables have p-values far above the conventional thresholds, suggesting a complete lack of explanatory power. Despite relatively large coefficient magnitudes in some cases, such as Price_Volatility (384.80) or Return_on_Equity (55.81), their lack of significance indicates that these estimates are unstable and likely driven by noise or overfitting in the data. While the model's AIC is very low (22) and residual variance close to zero, this does not imply quality; rather, it suggests a degenerate model with poor generalizability and overparameterization relative to the data. In summary, Model 4 does not provide meaningful insights and confirms that the target cluster lacks a clear financial signature that could be captured by the included variables.

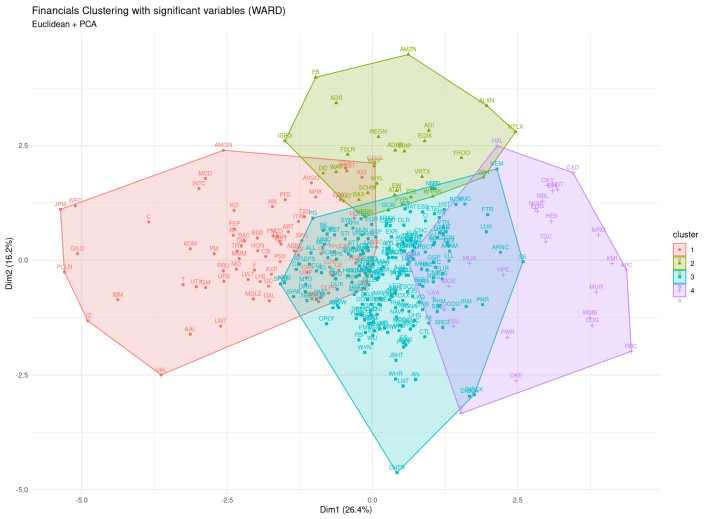
TABLE 12: Results of anova test on every variable and cluster, at a 5% significance level

Variable	Significancy
Quarterly_Return_Pct	TRUE
Price_Volatility	TRUE
Return_on_Equity	TRUE
Liquidity_CashRatio	TRUE
Cash_Flow_USD	FALSE
Net_Profit_USD	TRUE
EPS_USD	TRUE
PE_Ratio	TRUE
PB_Ratio	TRUE
MarketCapM	TRUE

To evaluate the global significance of the clustering structure, ANOVA tests were conducted for each financial variable across the four clusters. As shown in the table, 9 out of the 10 variables presented statistically significant differences at the 0.05 level. The only variable that did not reach significance was Cash_Flow_USD. All other variables — including Quarterly_Return_Pct, Price_Volatility, Return_on_Equity, Liquidity_CashRatio, Net_Profit_USD, EPS_USD, PE_Ratio, PB_Ratio, and MarketCapM — showed significant variation across the four groups.

Given the lack of significance of Cash_Flow_USD, we proceed by removing this variable from the dataset in order to assess whether the overall clustering structure improves as a result.

GRAPH 14: PCA projection with 4 clusters (Ward's method) using only significant variables



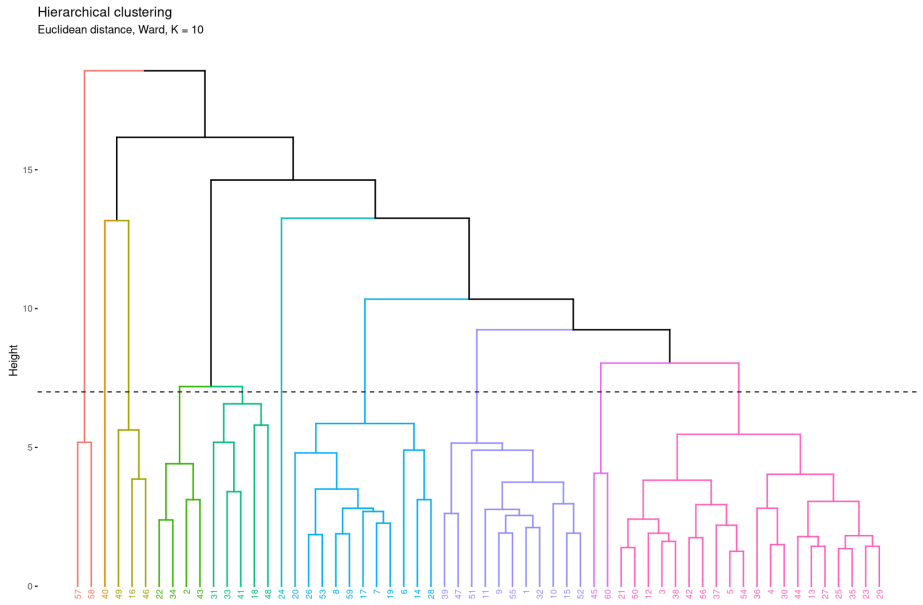
Graph 14 presents the clustering solution obtained using only the variables that were found statistically significant in previous ANOVA tests. Ward's hierarchical method was applied using Euclidean distance, and the resulting clusters are displayed in the space of the first two principal components, which now explain 26.4% and 16.2% of the total variance, respectively.

The clusters remain consistent in their general structure, with visible improvements in the compactness and separation of some groups. For instance, Clusters 2 and 4 show tighter and more distinct boundaries, while Cluster 3 continues to occupy a dense region in the centre. Although the overall clarity of the PCA plot appears slightly enhanced, this may result from the removal of a variable with limited discriminatory power.

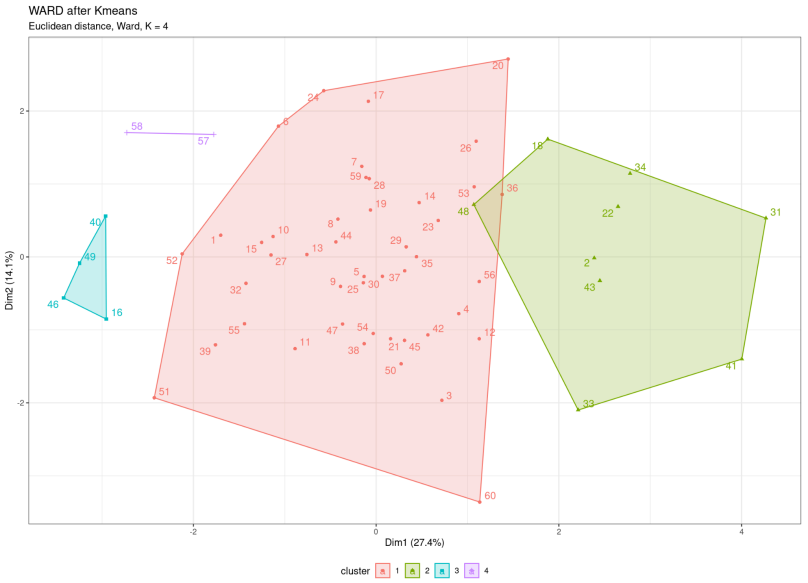
Nevertheless, the internal validation metrics in Table 1 show that Ward's method applied to only the significant variables performs worse than when applied to the broader filtered dataset. The Silhouette coefficient drops from 0.5854 to 0.4029, and the Dunn index decreases from 0.4029 to 0.2241, indicating weaker cohesion and separation. As a result, despite the visual neatness of this solution, it is not considered the definitive clustering structure due to its inferior quantitative performance.

GRAPH 15: Dendrogram with 60 clusters (Ward's method) and k=10

This dendrogram illustrates the hierarchical clustering structure based on Ward's method and Euclidean distance, ultimately dividing the dataset into 10 clusters. The height of the branches represents the dissimilarity between merged clusters, with higher merges indicating more distinct groups. This visualisation aids in determining an appropriate number of clusters and highlights the nested grouping relationships among the firms.



GRAPH 16: PCA projection with 4 clusters (Ward's method) after k-means clustering

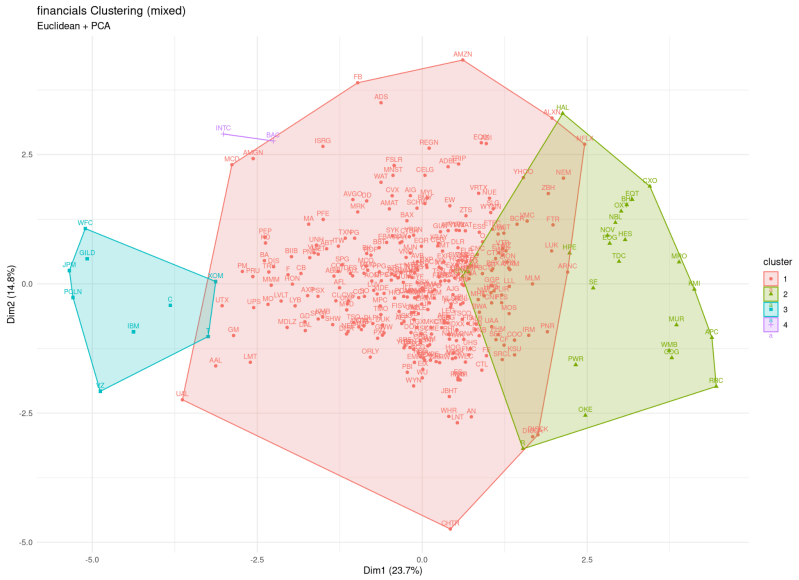


Graph 16 illustrates the results of a mixed clustering strategy. First, 60 micro-clusters were generated using the k-means algorithm. These initial clusters were then grouped into four macro-clusters using Ward's hierarchical clustering with Euclidean distance, producing the final structure shown in the plot. The PCA projection displays the clusters along the first two dimensions, which together explain 27.4% and 14.1% of the variance.

This two-step approach allows for capturing both local and global patterns in the data. The initial k-means step identifies fine-grained distinctions, while the hierarchical step aggregates those into broader, more interpretable groupings. The resulting clusters show good separation, with Clusters 1 and 2 containing the majority of the micro-clusters, and Clusters 3 and 4 forming more compact groupings. Notably, Cluster 4 contains only a few elements, suggesting a group of micro-clusters that are clearly distinct from the rest.

This method enhances flexibility by first capturing local structures and then consolidating them, offering a balance between granularity and interpretability.

GRAPH 17: PCA projection with 4 clusters (Ward's method) after k-means clustering, now using all observations



Graph 17 displays the result of the mixed clustering strategy, in which 60 initial k-means clusters were subsequently grouped into four macro-clusters using Ward's hierarchical method,now plotting all 330 individuals. The clusters are visualised in the PCA space of the first two components, explaining 23.7% and 14.8% of the variance, respectively.

Although the mixed approach aimed to combine the local sensitivity of k-means with the structure-preserving nature of hierarchical clustering, the outcome reveals a clear dominance of a single group (Cluster 1, in red), which contains the vast majority of firms. The remaining clusters are comparatively small, especially Cluster 4 (purple), which includes only a few firms. This severe imbalance is visually apparent in the graph and quantitatively supported by the distribution reported in Table 1 (299, 23, 9, 2).

The internal validation metrics further confirm the limitations of this method. The Silhouette coefficient is 0.3411, notably lower than both the hierarchical (0.5854) and k-means (0.4915) results on the same filtered dataset. Similarly, the Dunn index is 0.1275, compared to 0.4029 (hierarchical) and 0.2241 (k-means). The connectivity value could not be computed (NULL), limiting direct comparison on that metric.

In summary, despite the conceptual appeal of this two-step clustering strategy, the mixed method yields weaker quantitative results and produces highly unbalanced cluster sizes, leading to poorer overall performance on this dataset.

TABLE 13: Logistic regression applied to the Cluster 1 created with the mixed model

Mixed model	Estimate	Pr(>[z])	Significancy
(Intercept)	-83,89	0,00	***
Quarterly_Return_Pct	-0,01	0,77	
Price_Volatility	-0,90	0,38	
Return_on_Equity	0,59	0,05	.
Liquidity_CashRatio	1,37	0,00	***
Cash_Flow_USD	0,00	0,00	***
Net_Profit_USD	0,00	0,52	
EPS_USD	0,25	0,02	*
PE_Ratio	-5,66	0,00	***
PB_Ratio	0,00	0,94	
MarketCapM	3,82	0,00	***
Residual variance	95,20		
AIC	117,2		

Table 13 presents the logistic regression results for Cluster 1 obtained through the mixed clustering method. The model highlights the importance of several financial variables. Firms in this cluster are characterised by high liquidity, strong cash flow, lower PE ratios, and larger market capitalisations, all of which contribute clearly to cluster membership. Additionally, EPS and Return on Equity show positive associations, indicating a profitability profile linked to earnings quality and equity performance.

Other variables, such as Quarterly Return, Price Volatility, Net Profit, and PB Ratio, do not appear to have a relevant effect in this context.

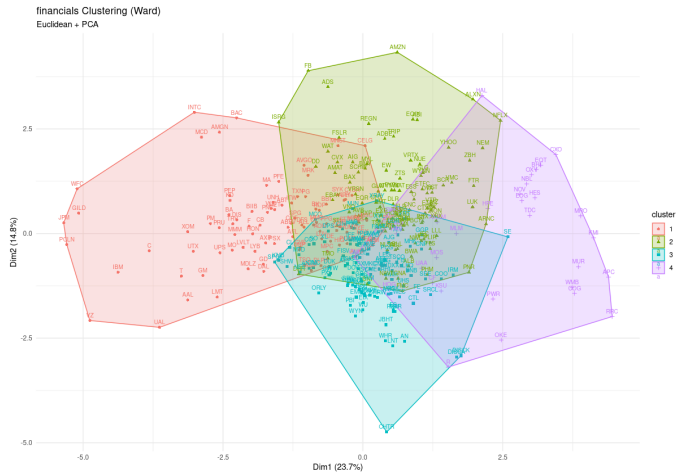
The model shows a strong overall fit, with a residual variance of 95.20 and an AIC of 117.2, very much in line with the results of Model 1.1. This confirms that the financial structure of Cluster 1 remains stable across different clustering approaches.

The results confirm that the clustering structure is consistent and well-supported across multiple analytical approaches. The PCA projections showed clear group separation, especially after filtering the dataset, where cluster compactness and visual clarity improved noticeably. These patterns were reinforced by ANOVA tests, which indicated significant differences across clusters for most financial variables. The comparison heatmaps between hierarchical and k-means clustering revealed strong alignment in the core groupings, though some mismatches at the cluster boundaries led to further analysis of overlapping segments.

The logistic regression models, applied to both hierarchical and mixed clustering results, confirmed the relevance of financial characteristics in defining each cluster. The addition of sector information improved the interpretability of certain clusters, particularly those with concentrated industry presence. Across all models, classification accuracy remained high, and AIC and deviance values indicated solid model fit. The consistency of results across clustering methods and variable sets suggests a robust and well-structured segmentation of firms in the dataset.

INTERPRETATION OF THE RESULTS

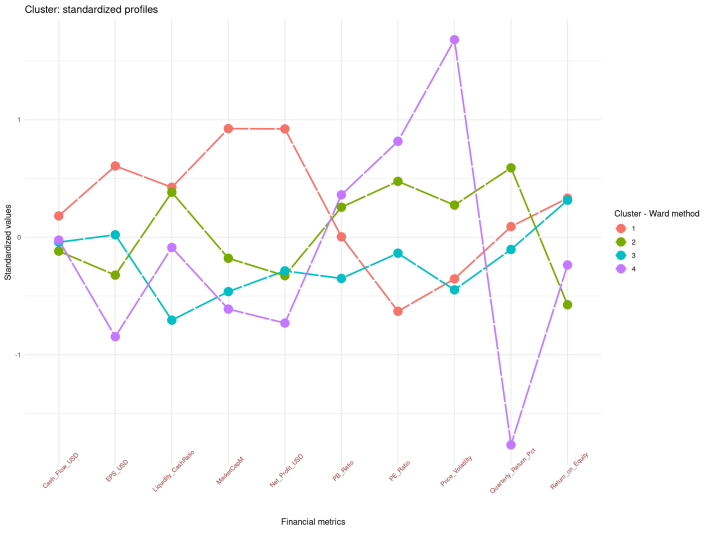
GRAPH 7: PCA projection with 4 clusters (Ward's method) removing graph 6's fourth cluster



GRAPH 18: Standardised profiles by cluster

The clustering process groups the companies into four distinct segments, each with a unique financial and sectoral profile. Before starting, when we talk about “negative or bad performance”, we refer to it as compared to the mean, not in absolute terms. As it is obvious, variables such as market cap can’t be negative.

Cluster 1 comprises firms with strong financial health. These companies display high profitability indicators—particularly ROE, EPS, and Net Profit—combined with low price volatility and solid liquidity. Their valuation levels are moderate, reflecting market confidence without excessive speculation. This profile aligns with mature, stable firms, often considered suitable for defensive



investment strategies. The sector distribution confirms a concentration in Financials and Industrials, industries typically associated with consistent performance.

Cluster 2 includes growth-oriented firms with higher risk. These companies show elevated PE and PB ratios, higher price volatility, and moderate earnings. Their financial profile suggests high market expectations, typical of firms in expansion or innovation-driven sectors. The cluster is dominated by Health Care and Technology firms, reinforcing the interpretation of a group with high potential returns but increased uncertainty.

Cluster 3 groups firms with a mixed financial profile. While these companies have low market capitalization, subdued earnings, and discounted valuations, they maintain relatively strong return on equity, suggesting lean or capital-efficient structures. Other profitability indicators, such as EPS and Net Profit, remain below average. The sector breakdown highlights the presence of Consumer Discretionary and Industrial firms, pointing to sensitivity to economic cycles and tighter margins.

Cluster 4 is defined by high volatility and sharply negative stock returns. Investor sentiment appears to weigh heavily on this group, which is overwhelmingly composed of Energy firms. The sharp decline in oil prices during the data year (2015) likely explains the discrepancy between core financials and market performance, having the worst quarterly return out of all clusters.

Together, these clusters reveal meaningful segmentation across financial performance and sector structure. The results show how unsupervised learning methods can uncover not only numerical distinctions but also strategic and sectoral patterns. This classification provides a valuable framework for portfolio construction, risk assessment, and peer comparison within and across groups.

Linking it with reality:

GRAPH 19: Crude oil price throughout 2015



These results observed across clusters are not isolated; they depend on macroeconomic conditions that defined the global economy in 2015. By understanding those conditions and our results, we can better understand why certain companies behaved the way they did.

Cluster 1 includes firms with high liquidity, strong cash flow, and large market capitalisations, most of which belong to the Financial and Industrial sectors. In 2015, central banks—particularly the European Central Bank and Bank of Japan—maintained expansionary monetary policies. In the U.S., the Federal Reserve held interest rates near zero until late in the year, contributing to a low-cost capital environment. These conditions supported asset valuations and investment activity, benefiting large, financially sound firms. Industrial companies profited from infrastructure and capital spending, while financials saw improved investor confidence and lower credit risk. This environment reinforces Cluster 1 as a group of low-risk, high-performance firms that thrived in a capital-friendly macroeconomic setting.

Cluster 2 is defined by high quarterly returns, elevated ROE, and strong price volatility. Its composition includes Technology, Health Care, Real Estate, and Financial firms, with almost no Energy companies. In 2015, tech and biotech stocks outperformed, driven by innovation optimism and investor risk appetite. The Nasdaq Composite and biotech indexes posted notable gains, reflecting

market enthusiasm for growth-oriented firms. Meanwhile, real estate investment trusts (REITs) attracted capital as low bond yields pushed investors toward dividend-yielding assets. The financial profile and sector makeup of this cluster point to firms that benefited from speculative inflows and growth momentum, albeit with increased volatility.

Cluster 3 shows firms with moderate net profits, smaller size, and subdued valuations, yet relatively strong ROE. It includes companies in the Industrial, Consumer Discretionary, and Utility sectors. The year 2015 was marked by global economic uncertainty, driven by a slowdown in China, fluctuating exchange rates, and commodity price instability. These pressures affected global trade volumes and manufacturing activity. Utilities provided stability but limited returns; industrials and consumer firms faced inconsistent demand. This cluster reflects companies operating in mature or cyclical sectors whose financial results were shaped by external demand constraints and margin pressure, despite decent equity returns.

Cluster 4 is dominated by Energy firms (53%) and is characterised by high volatility and sharply negative stock returns, despite otherwise mixed financial metrics. This aligns with the collapse in oil prices between 2014 and 2015, when WTI crude fell from over $100 to around $35 per barrel. The resulting revenue decline, asset write-downs, and funding pressures heavily affected energy company performance and stock valuations. This explains both the statistical unpredictability of this cluster and the sharp divergence from broader market trends. Companies in this group became difficult to model, reflecting the wider crisis in the energy sector.

These macro-financial patterns confirm that the cluster profiles are not only statistically distinct but also economically meaningful. They capture how firms responded to the structural and cyclical forces at play in 2015, offering a clearer understanding of corporate behaviour under specific market conditions. This contextual layer enhances the interpretive value of the clustering results and supports their relevance for financial analysis and investment strategy.

TABLE 14: Sector distribution throughout the 11 clusters

Sector	1	2	3	4	5	6	7	8	9	10	11
Financials	26	5	0	10	0	21	0	25	50	50	0
Industrials	24	5	0	23	14	12	12	0	0	0	0
Health Care	16	24	0	10	7	11	0	12	0	0	0
Consumer Staples	11	2	0	11	0	2	0	0	0	0	0
Information Technology	7	14	100	3	3	21	12	0	50	0	0
Consumer Discretionary	6	7	0	14	7	18	12	62	0	0	100
Energy	5	2	0	0	3	0	53	0	0	17	0
Materials	2	2	0	9	3	11	9	0	0	0	0
Real Estate	1	36	0	14	0	2	0	0	0	0	0
Telecommunications Serv	1	0	0	0	3	2	0	0	0	33	0
Utilities	1	2	0	6	59	2	0	0	0	0	0

TABLE 15: Sector distribution throughout the 11 clusters, as percentage of the cluster total

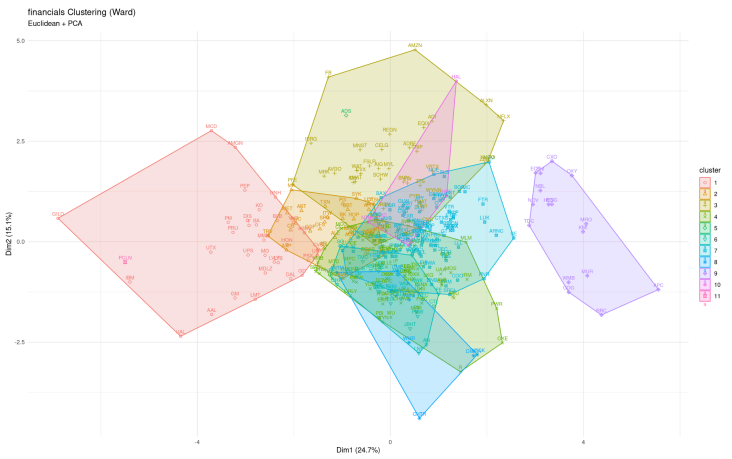
Sector	1	2	3	4	5	6	7	8	9	10	11
Financials	45	4	0	14	0	24	0	4	2	6	0
Industrials	38	4	0	30	8	13	8	0	0	0	0
Health Care	35	25	0	18	5	15	0	2	0	0	0
Consumer Staples	47	5	0	42	0	5	0	0	0	0	0
Information Technology	18	18	3	6	3	36	12	0	3	0	0
Consumer Discretionary	12	8	0	25	5	25	10	12	0	0	2
Energy	17	4	0	0	4	0	71	0	0	4	0
Materials	11	5	0	32	5	32	16	0	0	0	0
Real Estate	4	56	0	37	0	4	0	0	0	0	0
Telecommunications Serv	20	0	0	0	20	20	0	0	0	40	0
Utilities	4	4	0	17	71	4	0	0	0	0	0

A particularly relevant question in this analysis is whether the clusters identified are aligned with sector classification. At first glance, this would seem logical, as many financial ratios—such as volatility, cash flow, or valuation—are known to vary systematically across sectors. However, the

results suggest a different dynamic. As observed across several models and visualisations, sector is generally not a significant determinant of cluster membership, with only a few exceptions.

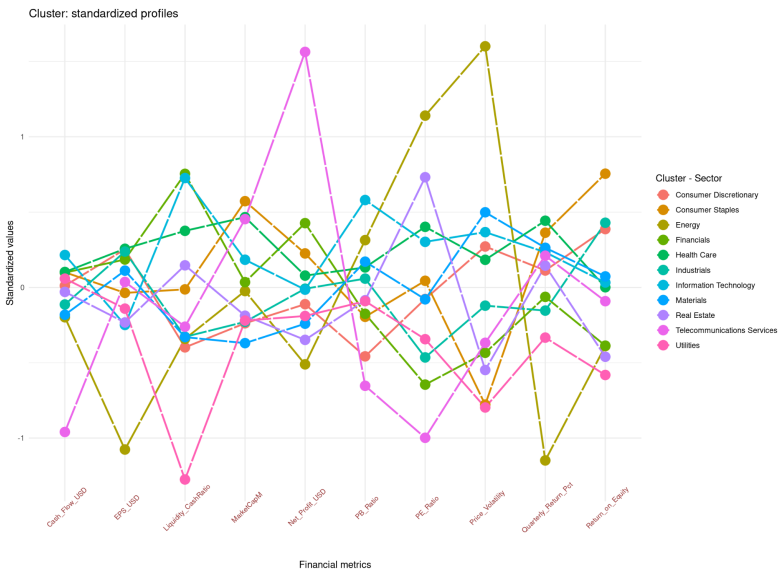
The clearest evidence of this is seen in the sector-cluster contingency tables, where firms from the same sector are often distributed across multiple clusters. This indicates that clustering is more influenced by performance-based variables, such as profitability, market cap, and volatility, than by sectoral affiliation.

GRAPH 20: PCA projection with 11 clusters (Ward's method)



The standardised profile chart by sector shows that within-sector variation is high—sectoral curves cross repeatedly and do not remain above or below the average across all metrics. This suggests that firms within the same sector can differ meaningfully across the financial dimensions used in the clustering.

GRAPH 21: Standardised profiles by sector



Moreover, the attempt to match the number of clusters to the number of sectors (11 in both cases) does not result in clusters that are sector-specific. In fact, some clusters (e.g., Cluster 3 or Cluster 10) are

highly dominated by a single sector, while others (e.g., Clusters 1, 2, 4, 6) are clearly sectorally heterogeneous. This heterogeneity confirms that the clustering is driven more by shared financial behaviours than by shared industrial classification.

In summary, firms tend to be grouped more strongly by their financial performance patterns than by their sector. While some sectors (such as Energy or Real Estate) do exhibit internal homogeneity and cluster dominance, most sectors are dispersed. This reflects a key insight: in 2015, firm behaviour—especially under stress or growth conditions—cut across sector boundaries, highlighting the importance of dynamic, data-driven clustering over static classifications.

CONCLUSIONS

When we started this analysis, we had the aim of segmenting a sample of 340 companies based on their financial structures and behaviour. In order to do this, we conducted a pre-processing of the dataset, transforming skewed variables, removing the outliers and ensuring the standardisation of the dataset for clustering. This allows us to treat the data homogeneously, ignoring the scale and variability of the original variables.

Different clustering methods were used and evaluated, in this case, Ward and K-means. The most robust of the two was the Ward method, across all metrics as we expected, principally due to the small size of the dataset. To identify the number of clusters, we used the Elbow Method. Validation metrics (connectivity, silhouette width, Dunn index) were used to identify which of the methods offered more robust and consistent results. ANOVA and logistic regressions were used to identify which variables were relevant and the results showed that almost all variables were significant for cluster separation, with the exception of the cash flow variable, which was the only non-significant variable and was excluded in some analyses and included in others. We obtained mixed results across models, with some of them being better with the sectors and others where they weren't significant, performed better without them.

The clusters obtained are not only statistically distinct, but analysing their companies reveals distinct financial profiles, as we indicated in the interpretation.

We then explored the relationship between the different clusters and the original categorical variables. After conducting the analysis, we found that some clusters showed a strong relation with particular sectors, such as having the energy companies concentrated in a single group. However, most cluster companies are from different sectors, confirming that the financial structure of the companies does not always match their sector. This highlights the importance of statistical clustering over traditional methods such as grouping by sector.

In summary, we successfully segmented the dataset into four interpretable and statistically robust groups. Mainly: Cluster 1 includes firms with high profitability, low volatility, and large size; Cluster 2 is defined by high valuations, high volatility, and poor earnings; Cluster 3 groups firms with low valuation, strong ROE, and small size; and Cluster 4 is characterized by high volatility, negative returns, and a concentration in the energy sector. The clustering reflects the differences in the quantitative characteristics of the companies and not only in the sector they belong to. This clustering may serve as a tool for investors, analysts or corporate strategists. These groupings can support portfolio construction, benchmarking, and the detection of investment opportunities based on underlying financial patterns rather than superficial categorisations.

AI Usage Statement

In this assignment, AI tools have been used in the following ways:
ChatGPT was used to assist with the coding in R, optimising some functions and helping to debug some errors, and played an important role in order to interpret the results and their implications.